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Modelling and Reformulating Constraint Satisfaction Problems:

Towards Systematisation and Automation

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Foreword

Many organisations have scheduling, assignment, supply chain and other problems that could be solved with a constraint programming toolkit. Although the solution of these problems is of vital importance, constraint programming toolkits are not widely used because there is insufficient expertise available to model problems as constraint programs.

This so-called modelling bottleneck could be reduced by the development of a general, principled understanding of modelling that in the future could guide the manual or automatic formulation of models and the choice among alternative models.

Researchers and practitioners have developed effective models for a wide range of problems. The time has come to form generalisations from these case studies that can be used to guide modelling in the future. These generalisations could then be systematised for use by a non-expert and be codified in textbooks in much the same way that data structuring expertise is. Ultimately this modelling expertise could be embedded in automated modelling tools. Progress on any of these fronts would bring the proven power of constraint programming to a wider user base.

This “Third International Workshop of Modelling and Reformulating Constraint Satisfaction Problem: Towards Systematisation and Automation” was convened to provide a forum for researchers who share these goals.

This volume contains nine contributed papers and one invited paper from the invited speaker, Michael Trick.

We wish to thank all the authors who submitted papers to this workshop; the members of the programme committee; the invited speaker, Michael Trick; and the CP’2004 Tutorial and Workshop Chair, Barry OSullivan.

August 2004

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